

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE

SUBJECT : R-PROGRAMMING

PRACTICAL13

AIM : Performing linear regression analysis using lm() (R).

OUTPUT

The screenshot displays the RStudio interface with the following content:

```
R - R4.5.1 - ~/...
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins

Source
Console Terminal Background Jobs

> # 2. Load dataset
> insurance <- read.csv("insurance.csv")
>
> # 3. View dataset
> head(insurance)
  age  sex  bmi children smoker  region  charges
1  19 female 27.900      0    yes southwest 16884.924
2  18  male 33.770      1    no  southeast 1725.552
3  28  male 33.000      3    no  southeast 4449.462
4  33  male 22.705      0    no northwest 21984.471
5  32  male 28.880      0    no northwest 3866.855
6  31 female 25.740      0    no  southeast 3756.622
> str(insurance)
'data.frame':   1338 obs. of  7 variables:
 $ age      : int  19 18 28 33 32 31 46 37 37 60 ...
 $ sex      : chr  "female" "male" "male" "male" ...
 $ bmi      : num  27.9 33.8 33 22.7 28.9 ...
 $ children : int  0 1 3 0 0 0 1 3 2 0 ...
 $ smoker   : chr  "yes" "no" "no" "no" ...
 $ region   : chr  "southwest" "southeast" "southeast" "northwest" ...
 $ charges  : num  16885 1726 4449 21984 3867 ...
> summary(insurance)
   age      sex      bmi      children      smoker      region      charges
Min.   :18.00   Length:1338   Min.   :15.96   Min.   :0.000   Length:1338
1st Qu.:27.00   Class :character   1st Qu.:26.30   1st Qu.:0.000   Class :character
Median :39.00   Mode  :character   Median :30.40   Median :1.000   Mode  :character
Mean   :39.21
3rd Qu.:51.00
Max.   :64.00
   region      charges
Length:1338   Min.   : 1122
Class :character 1st Qu.: 4740
Mode  :character Median : 9382

Max.   :63770
> # 4. Convert categorical variables to factors
> insurance$sex <- factor(insurance$sex)
> insurance$smoker <- factor(insurance$smoker)
> insurance$region <- factor(insurance$region)
>
> # 6. Simple Linear Regression
> simple_model <- lm(charges ~ age, data = insurance)
> summary(simple_model)

Call:
lm(formula = charges ~ age, data = insurance)

Residuals:
    Min       1Q   Median       3Q      Max
-8059   -6671   -5939    5440   47829

Coefficients:
(Intercept)    3165.9      937.1    3.378  0.000751 ***
age           257.7      22.5   11.453  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11560 on 1336 degrees of freedom
Multiple R-squared:  0.08941, Adjusted R-squared:  0.08872
F-statistic: 131.2 on 1 and 1336 DF, p-value: < 2.2e-16

>
> # 7. Multiple Linear Regression
> multiple_model <- lm(charges ~ age + bmi + children + sex + smoker + region,
+ data = insurance)
```

The Environment pane on the right shows the loaded objects: insurance (1338 obs. of 7 variables), logistic_m (Large glm (30 elements, 65...)), multiple_m (List of 13), and simple_mod (List of 12). The Files pane shows the project files, including insurance.csv (543 KB).

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The image displays two screenshots of the RStudio interface, showing the execution of an R script for multiple linear regression and diagnostic plots.

Top Screenshot:

- Source:** The script defines a multiple linear regression model using the `lm` function on the `insurance` dataset. The model includes predictors: `age`, `bmi`, `children`, `sex`, `smoker`, and `region`. The response variable is `charges`.
- Console:** The output shows the residuals, coefficients, and summary statistics for the model. The residuals are centered around zero, and the coefficients are statistically significant.
- Environment:** The `insurance` dataset is loaded, containing 1338 observations and 7 variables. The `multiple_model` object is also present.

Bottom Screenshot:

- Source:** The script continues with diagnostic plots and prediction. It uses `par(mfrow = c(2,2))` to display four diagnostic plots (residuals vs fitted, Q-Q plot, residuals vs leverage, and Cook's distance). It then predicts the charges for the `insurance` dataset using the `predict` function.
- Console:** The output shows the predicted charges for the `insurance` dataset. The predicted values are displayed as a vector of 1338 values.
- Environment:** The `insurance` dataset and the `multiple_model` object are still present. The `predicted_charges` object is also visible.

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The image displays two screenshots of the RStudio environment, showing the results of two different linear models.

Top Screenshot: multiple_model

The Environment pane shows the following variables:

Name	Type	Value
multiple_model	list [13] (S3: lm)	List of length 13
coefficients	double [9]	-11939 257 339 476 -131 23849 ...
residuals	double [1338]	-8408.8 -1723.1 -2257.5 18229.6 -1725.6 36.8 ...
effects	double [1338]	-485415 132402 -73800 23906 -24134 350529 ...
rank	integer [1]	9
fitted.values	double [1338]	25294 3449 6707 3755 5592 3720 ...
assign	integer [9]	0 1 2 3 4 5 ...
qr	list [5] (S3: qr)	List of length 5
df.residual	integer [1]	1329
contrasts	list [3]	List of length 3
xlevels	list [3]	List of length 3
call	language	lm(formula = charges ~ age + bmi + children + sex + smoker + region, data = ...
terms	formula	charges ~ age + bmi + children + sex + smoker + region
model	list [1338 x 7] (S3: data.frame)	A data.frame with 1338 rows and 7 columns

The Console shows the following commands and output:

```
> r_squared <- cor(predicted_charges, insurance$charges)^2
> r_squared
[1] 0.750913
> View(multiple_model)
```

Bottom Screenshot: simple_model

The Environment pane shows the following variables:

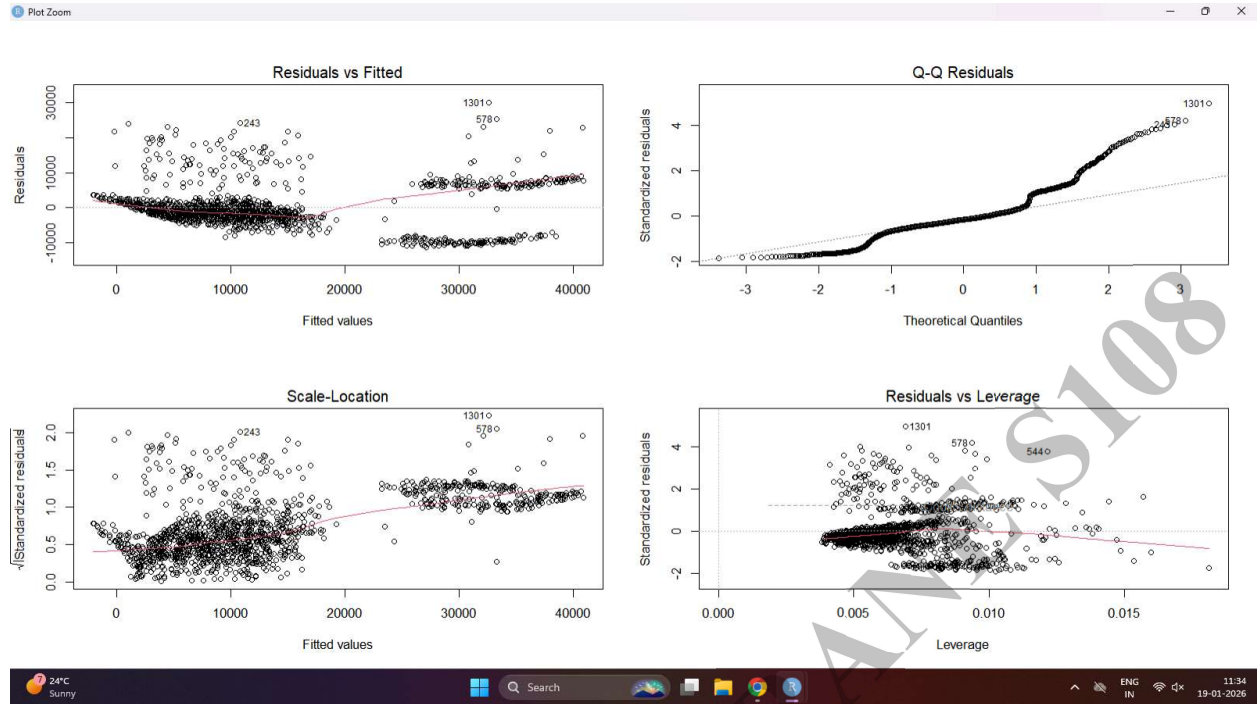
Name	Type	Value
simple_model	list [12] (S3: lm)	List of length 12
coefficients	double [2]	3166 258
residuals	double [1338]	8822 -6079 -5933 10314 -7546 -7399 ...
effects	double [1338]	-485415 132402 -6041 10146 -7702 -7543 ...
rank	integer [1]	2
fitted.values	double [1338]	8063 7805 10382 11671 11413 11155 ...
assign	integer [2]	0 1
qr	list [5] (S3: qr)	List of length 5
df.residual	integer [1]	1336
xlevels	list [0]	List of length 0
call	language	lm(formula = charges ~ age, data = insurance)
terms	formula	charges ~ age
model	list [1338 x 2] (S3: data.frame)	A data.frame with 1338 rows and 2 columns

The Console shows the following commands and output:

```
> r_squared
[1] 0.750913
> View(multiple_model)
> View(simple_model)
```

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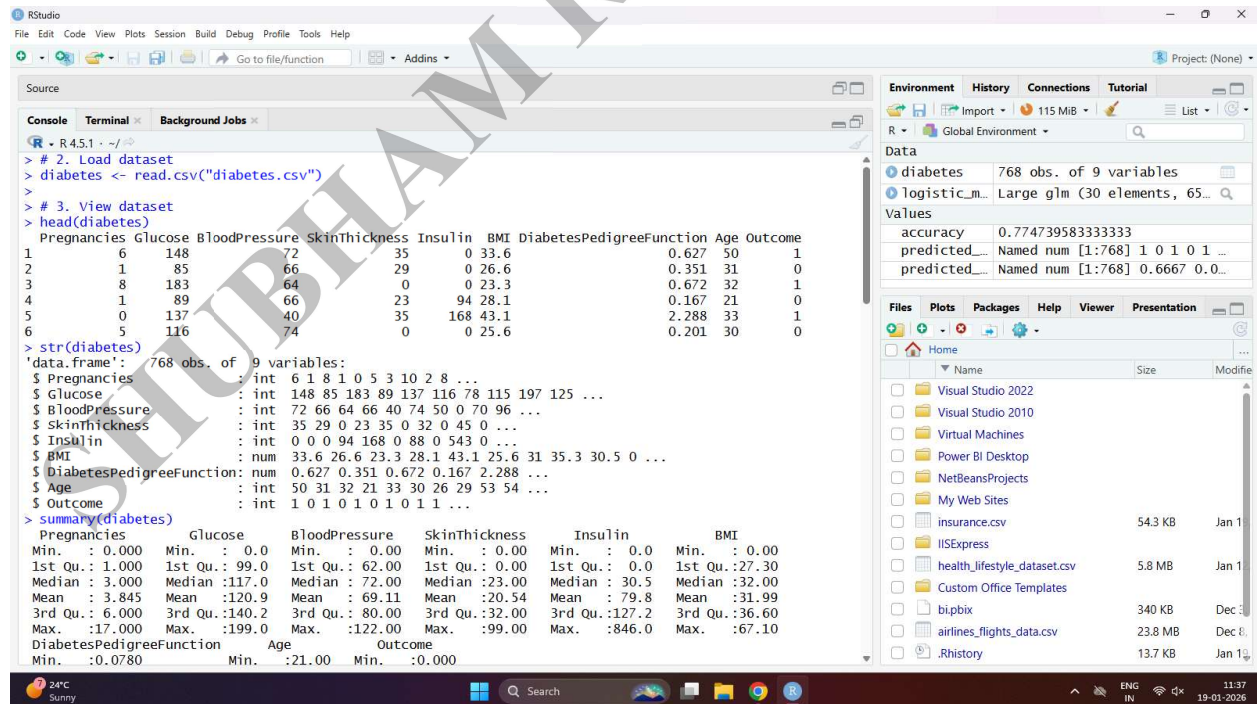
SUBJECT : R-PROGRAMMING



PRACTICAL14

AIM : Performing logistic regression using glm() (R).

OUTPUT



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SUBJECT : R-PROGRAMMING

The image displays two screenshots of the RStudio interface, showing the execution of R code for a logistic regression model.

Top Screenshot:

- Source:** Shows the R script with comments and code for data preparation and model fitting.
- Console:** Displays the output of the code, including summary statistics for the 'diabetes' dataset and the results of the logistic regression model.
- Environment:** Shows the loaded objects: 'diabetes' (768 obs. of 9 variables) and 'logistic_m...' (Large glm (30 elements, 65...)).
- Files:** Shows the file explorer with various files and folders.

Bottom Screenshot:

- Source:** Shows the R script with comments and code for data preparation and model fitting.
- Console:** Displays the output of the code, including the results of the logistic regression model, coefficients, and odds ratios.
- Environment:** Shows the loaded objects: 'diabetes' (768 obs. of 9 variables) and 'logistic_m...' (Large glm (30 elements, 65...)).
- Files:** Shows the file explorer with various files and folders.

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SUBJECT : R-PROGRAMMING

The image displays two screenshots of the RStudio interface, showing the results of a logistic regression analysis.

Top Screenshot: The console shows the output of the `glm` function, including deviance, AIC, and coefficients. The environment pane shows the `diabetes` data frame and the `logistic_model` object.

```
Null deviance: 993.48 on 767 degrees of freedom
Residual deviance: 734.99 on 762 degrees of freedom
AIC: 746.99

Number of Fisher Scoring iterations: 5

> # 9. Odds Ratios
> exp(coef(logistic_model))
      (Intercept)  Pregnancies  Glucose BloodPressure      BMI      Age
0.00035004      1.12614761    1.03436446    0.98657715    1.09522613    1.01723334

> # 10. Predict probabilities
> predicted_prob <- predict(logistic_model, type = "response")

> # 11. Convert probabilities to class labels
> predicted_class <- ifelse(predicted_prob > 0.5, 1, 0)

> # 12. Confusion Matrix
> table(Predicted = predicted_class, Actual = diabetes$Outcome)
      Actual
Predicted 0  1
         0 437 110
         1  63 158

> # 13. Model Accuracy
> accuracy <- mean(predicted_class == diabetes$Outcome)
> accuracy
[1] 0.7747396
> View(logistic_model)
```

Bottom Screenshot: The environment pane shows the `logistic_model` object, which is a list of 30 elements. The console shows the output of the `summary(logistic_model)` function, providing a detailed overview of the model's components, including coefficients, residuals, fitted values, effects, R-squared, rank, QR, family, linear predictors, deviance, AIC, null deviance, iter, weights, prior weights, df.residual, df.null, y, and converged.

```
logistic_model list [30] (S3: glm, lm) List of length 30
 coefficients double [6] -7.9575 0.1188 0.0338 -0.0135 0.0910 0.0171
 residuals double [768] 1.50 -1.05 1.38 -1.06 1.54 -1.20 ...
 fitted.values double [768] 0.6667 0.0517 0.7266 0.0569 0.6491 0.1676 ...
 effects double [768] 5.352 4.033 -10.385 0.906 -6.216 1.863 ...
 R double [6 x 6] -10.96 0.00 0.00 0.00 0.00 -46.22 37.47 0.00 ...
 rank integer [1] 6
 qr list [5] (S3: qr) List of length 5
 family list [13] (S3: family) List of length 13
 linear.predictors double [768] 0.693 -2.909 0.977 -2.809 0.615 -1.603 ...
 deviance double [1] 734.9861
 aic double [1] 746.9861
 null.deviance double [1] 993.4839
 iter integer [1] 5
 weights double [768] 0.2222 0.0490 0.1987 0.0536 0.2278 0.1395 ...
 prior.weights double [768] 1 1 1 1 1 ...
 df.residual integer [1] 762
 df.null integer [1] 767
 y double [768] 1 0 1 0 1 ...
 converged logical [1] TRUE
```

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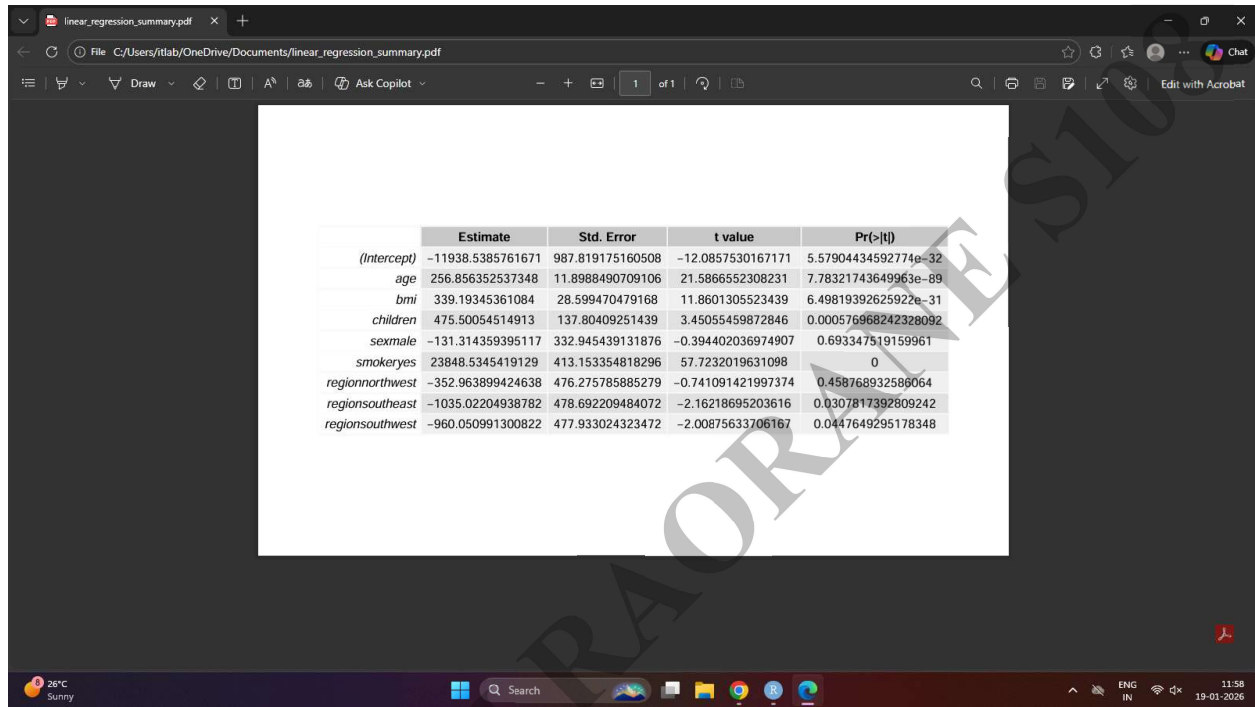
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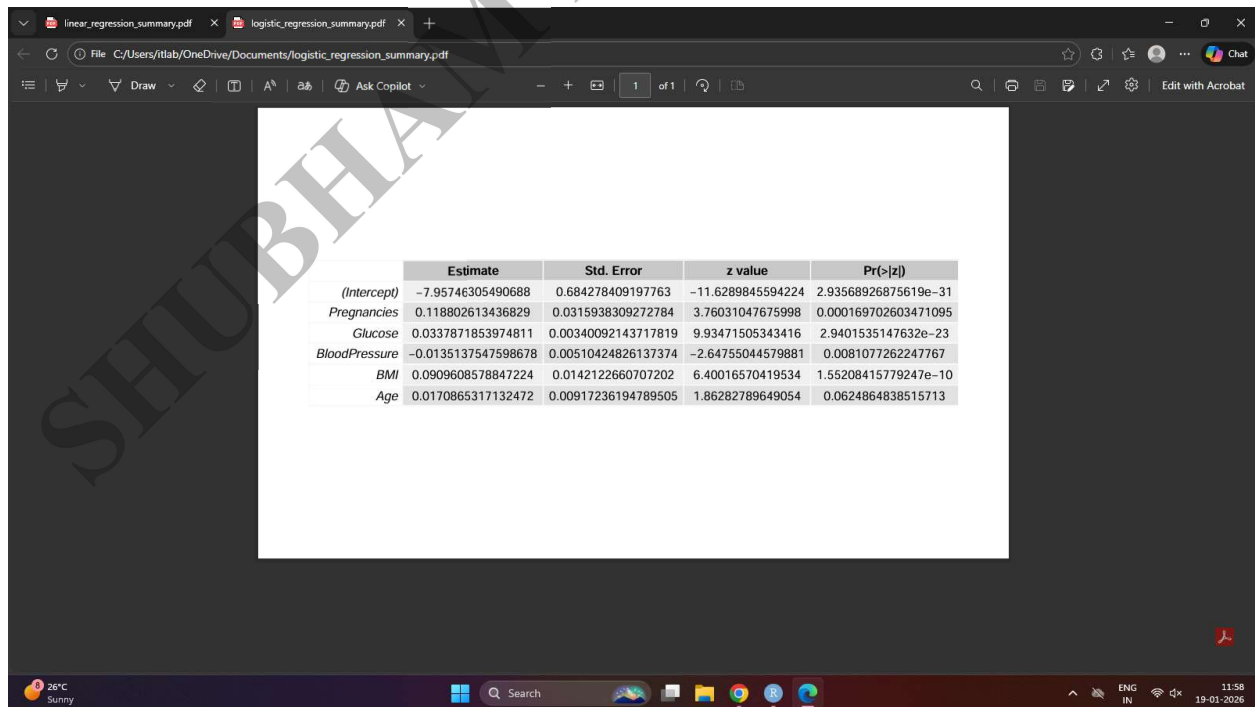
AIM : Exporting results into external files (Excel, CSV, PDF) using `write.csv()` and `writexl (R)`.

OUTPUT



The screenshot shows a PDF viewer window titled 'linear_regression_summary.pdf'. The document content is a table with 5 columns: (Intercept), Estimate, Std. Error, t value, and Pr(>|t|). The table lists coefficients for various variables including age, bmi, children, sexmale, smokeryes, regionnorthwest, regionsoutheast, and regionsouthwest. A large diagonal watermark 'SHUBHAM RAORANE S108' is visible across the page.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-11938.5385761671	987.819175160508	-12.0857530167171	5.57904434592774e-32
age	256.856352537348	11.8988490709106	21.5866552308231	7.78321743649963e-89
bmi	339.19345361084	28.599470479168	11.8601305523439	6.49819392625922e-31
children	475.50054514913	137.80409251439	3.45055459872846	0.000576968242328092
sexmale	-131.314359395117	332.945439131876	-0.394402036974907	0.693347519159961
smokeryes	23848.5345419129	413.153354818296	57.7232019631098	0
regionnorthwest	-352.963899424638	476.275785885279	-0.741091421997374	0.458768932586064
regionsoutheast	-1035.02204938782	478.692209484072	-2.16218695203616	0.0307817392809242
regionsouthwest	-960.050991300822	477.933024323472	-2.00875633706167	0.0447649295178348



The screenshot shows a PDF viewer window titled 'logistic_regression_summary.pdf'. The document content is a table with 5 columns: (Intercept), Estimate, Std. Error, z value, and Pr(>|z|). The table lists coefficients for variables including Pregnancies, Glucose, BloodPressure, BMI, and Age. A large diagonal watermark 'SHUBHAM RAORANE S108' is visible across the page.

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-7.95746305490688	0.684278409197763	-11.6289845594224	2.93568926875619e-31
Pregnancies	0.118802613436829	0.0315938309272784	3.76031047675998	0.000169702603471095
Glucose	0.0337871853974811	0.00340092143717819	9.93471505343416	2.9401535147632e-23
BloodPressure	-0.0135137547598678	0.00510424826137374	-2.64755044579881	0.0081077262247767
BMI	0.0909608578847224	0.0142122660707202	6.40016570419534	1.55208415779247e-10
Age	0.0170865317132472	0.00917236194789505	1.86282789649054	0.0624864838515713

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